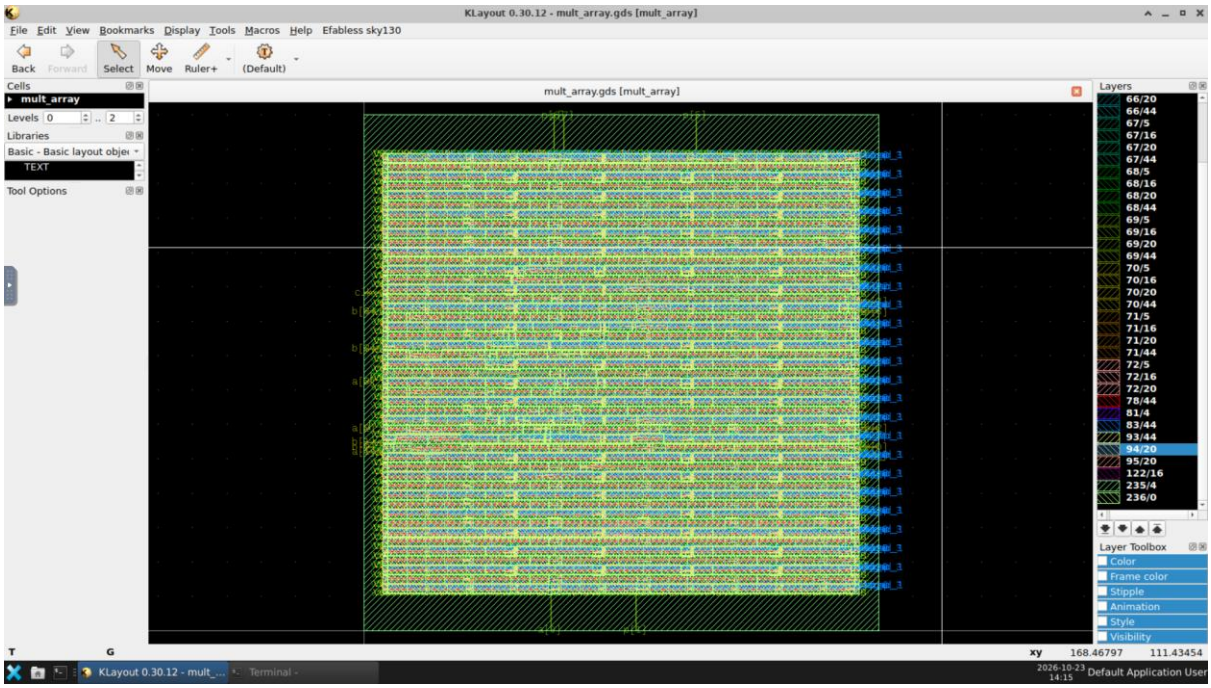
PART D

Results Table

Metric	Value
CLOCK_PERIOD used (ns)	**6.5**
Number of runs needed	**3** on SKY130 (3.5, 6.0 and 6.5 ns), plus one earlier attempt with the wrong PDK (IHP) that is not counted
Core area (µm²)	**17759.5** (die area 22500 = 150 × 150)
Cells after synthesis (from Part B)	**63**
Cells after place and route	**1901**
Kinds of cells added	Tap cells (245), decap cells (1296), fill cells (270), port buffers (17 buf_2), clock-tree buffers (3 clkbuf_16), delay cells (2 clkdlybuf4s50_1)
Wirelength (µm)	**2405**
Setup slack (ns)	**+0.085** (worst corner, max_ss_100C_1v60)
Post-route Fmax (MHz)	1000 / (6.5 − 0.085) = **155.9**
Fmax from Part C (MHz)	**383.1**
DRC errors	**0** (Magic 0, KLayout 0)
LVS errors	**0**

Layout Screenshot 1:



Layout Screenshot 2:

